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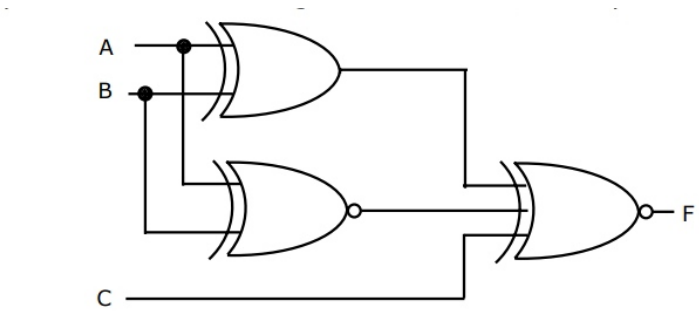
Date: 20 FEB 2026

GATE 2010 EC - Question 12 Analysis

Question 12

For the output F to be 1 in the logic circuit shown, the input combination should be:



A. A=1, B=1, C=0

B. A=1, B=0, C=0

C. A=0, B=1, C=0

D. A=0, B=0, C=1

Question Analysis

- Two input XOR gate: $X = A \oplus B$
- Two input XNOR gate: $Y = A \odot B$
- Three input XNOR gate: $F = X \odot Y \odot C$
- Final expression:

$$F = (A \oplus B) \odot (A \odot B) \odot C$$

The Truth Table

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Hardware Implementation

The circuit is implemented using Arduino UNO board. Inputs A, B, C are given through digital pins. Output F is verified using LED and seven segment display.

Required Components

S.No	Component
1	Arduino Uno Board
2	Breadboard
3	Seven Segment Display
4	220Ω Resistors
5	Jumper Wires
6	USB Cable

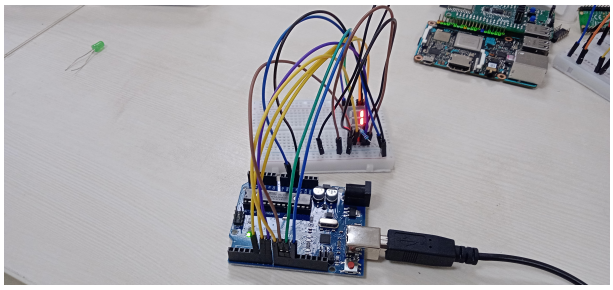
Pin Connections

Component	Arduino Pin
Input A	Digital 2
Input B	Digital 3
Input C	Digital 4
LED Output F	Digital 12

Code Uploading Steps

1. Create PlatformIO project
2. Write code in main.cpp
3. Run command: pio run
4. Copy generated .hex file
5. Connect Arduino using OTG
6. Upload using "Upload Precompiled"
7. Verify output

Experimental Output



Conclusion

From the truth table, $F = 1$ whenever $C = 1$. Hence the correct option is:

$$\boxed{\text{Option (D)}}$$

The hardware implementation confirms the theoretical logic.